



MATSOC

PRESENTS

AlgoAscent: Competitive Programming

Unlock the Art of Problem Solving

Mentored By

Aryan Chaturvedi

Tanay Pathode

Ayush Sharma

Project Overview: Your CP Launchpad

What is Competitive Programming

Competitive Programming is a mind sport where participants need to write programs to solve well-defined problems

These problems are often related to data structures, algorithms, maths and logic

 **2 Sessions / Week** (1 hr each)

Some Popular Programming Sites



The 'Why' of CP

Competitive Programming is the *Gold Standard* for mastering Data Structures and Algorithms (DSA) and developing problem-solving skills.

★ *Career Prep:* Master the core DSA required for tech interviews.

💓 Mind Sport: It's genuinely fun and highly addictive (in a good way!).

👥 Unlock the Community: Finally understand the famous memes and discussions on Codeforces and AtCoder :)

The AlgoAscent: Weekly Plan

Wk
1

Foundations, STL

Basic Project Intro/Flow, IDE Setup,
Full STL Usage & Understanding
TC/SC/ASC Recursion
Binary Search and Initial sorting
methods $O(N^2)$, Count Sort, Radix
Sort Intro)

Wk
2

Number Theory I & II

Primality testing, Binary operations,
and Bitwise operations (including
Binary Exponentiation)
Modular arithmetic, Prefix Sums,
and basic approaches like
Complete search

Wk
3

Optimization Tec, Greedy

Two-pointer method and the
Sliding Window technique.
Core principles of Greedy
Algorithms (Including classic
problems)

Wk
4

Dynamic Programming

Introduction to DP: State definition,
recurrence relations, and solving
fundamental DP patterns
Solving advanced DP problems and
recognizing complex transition
patterns

Wk
5

Graph Theory I & II

Graph terminologies, Trees, and
basic traversal algorithms: DFS and
standard BFS
Advanced BFS variants (0-1 BFS,
multisource), and Dijkstra's shortest
path algorithm

Wk
6

Advanced Data Structures

Lowest Common Ancestor (LCA),
Segment Trees, and Sparse Tables
for range queries
Introduction to Interactive
Programming and advanced
problem-solving techniques

Logistics & Competitive Evaluation

STRUCTURE

Duration	6 Weeks (1.5 Months)
Frequency	2 Sessions / Week (1 Hour each)
Workload	~20 hours/week (Practice)
Difficulty	Medium (Motivated Beginners)
Mode	Online

Competitive Evaluation

INTERNAL CHALLENGES

Biweekly Contests

Test your skills against your peers.

MANDATORY EVALUATION

Codeforces Contests

Gain real-time rating and experience essential for progress.

Prerequisites

- Must have done ESC111 and ESC112
- Basic C++ & Time Complexity Analysis
- Must be familiar with STL Basics

(We build CP skills from this foundation.)

Project Mentors



Ayush Sharma
ayushs23@iitk.ac.in
[7999619869]



Aryan Chaturvedi
aryanc23@iitk.ac.in
[8738932371]



Tanay Pathode
tanayp23@iitk.ac.in
[8446445740]